

Heuristic Evaluation — Results

Nielsen's 10 Heuristics | HCI Evaluation | Evaluators: M1, M2, M3, M4 | 05/06/2026

1. Overview

All four team members (M1, M2, M3, M4) conducted a heuristic evaluation of the AvatarFit prototype, evaluating the live application.

Each interface element and interaction flow was assessed against Nielsen's 10 usability heuristics. Issues were assigned a severity score from 0 (not a problem) to 4 (catastrophic). Results were consolidated into a single report by M1.

Severity scale used:

Score	Label	Meaning
0	Not a problem	No usability issue.
1	Cosmetic	Fix only if time permits.
2	Minor	Low priority; user can work around it.
3	Major	Important to fix; causes significant difficulty.
4	Catastrophic	Must be fixed; prevents task completion.

2. Results per Heuristic

H1 — Visibility of System Status

Finding	Screen / Task	Severity
The system clearly shows when the webcam is active and recording.	<i>Warm-up screen</i>	0 — OK
Visual feedback (red/green skeleton) clearly communicates form status in real time.	<i>Workout screen</i>	0 — OK

H2 — Match Between System and the Real World

Finding	Screen / Task	Severity
Exercise names (squat, lunge, curl, plank) are correct and familiar to users.	<i>Exercise selection</i>	0 — OK

Correction messages could be slightly more specific in their wording.	<i>Workout screen</i>	1 — Cosmetic
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H3 — Freedom and Controls by the Users

Finding	Screen / Task	Severity
User can return to the home screen easily at any time.	<i>All screens</i>	0 — OK
Rep counter can be reset during a session.	<i>Workout screen</i>	0 — OK

H4 — Coherence and Standards

Finding	Screen / Task	Severity
Colours are used consistently throughout: red always indicates bad form, green/yellow indicates correct form.	<i>Workout screen</i>	0 — OK

H5 — Error Prevention

Finding	Screen / Task	Severity
No on-screen instruction tells the user the correct starting position or that certain exercises require a sideways stance.	<i>Starting an exercise</i>	3 — Major
No guidance is provided to help the user check whether the webcam is correctly positioned before starting.	<i>Before exercise</i>	3 — Major

H6 — Recognition Rather Than Recall

Finding	Screen / Task	Severity
All controls and options are persistently visible on screen.	<i>All screens</i>	0 — OK
No information from a previous screen needs to be remembered to complete a task.	<i>All screens</i>	0 — OK

H7 — Flexibility and Efficiency of Use

Finding	Screen / Task	Severity
Voice commands are available and functional, though not prominently advertised.	<i>Workout screen</i>	0 — OK

Users can switch exercise directly from the workout screen without returning to the home screen.	<i>Workout screen</i>	0 — OK
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H8 — Minimalistic Design

Finding	Screen / Task	Severity
The workout screen contains no irrelevant information; all elements serve a clear purpose.	<i>Workout screen</i>	0 — OK
The rep counter is clearly visible and easy to read at a glance.	<i>Workout screen</i>	0 — OK

H9 — Helping the Users Recognize, Diagnose and Fix the Errors

Finding	Screen / Task	Severity
The app now provides specific verbal corrections describing the required adjustment (e.g. 'lower your hips').	<i>Workout screen</i>	0 — OK

H10 — Guides and Documentation

Finding	Screen / Task	Severity
There is currently no tutorial or onboarding flow for first-time users.	<i>First use</i>	2 — Minor
Voice commands are discoverable via a button on the workout screen.	<i>Workout screen</i>	0 — OK

3. Consolidated Issue Summary

All issues ranked by severity (highest first):

#	Issue	Heuristic	Severity
1	No starting position or orientation instruction before exercises.	<i>H5 — Error Prevention</i>	3 — Major
2	No webcam positioning guidance before starting.	<i>H5 — Error Prevention</i>	3 — Major
3	No tutorial or onboarding for first-time users.	<i>H10 — Guides and Documentation</i>	2 — Minor

4	Correction messages could be more specific in wording.	H2 — Match with Real World	1 — Cosmetic
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4. Key Findings

OK App scores well across 8 out of 10 heuristics No violations were found for H1, H3, H4, H6, H7, H8, or H9. H2 had one cosmetic-level issue. The majority of the interface design is solid and does not cause usability problems.

(!) Two major issues found under H5 (Error Prevention) The most significant finding is the absence of any initial guidance about exercise starting position and webcam placement. Both were rated severity 3 (Major). These are pre-task issues that could prevent users from getting useful feedback if they start in the wrong position.

(!) No tutorial or onboarding (H10, severity 2) First-time users receive no introduction to the app's key concepts (skeleton feedback, voice commands, exercise switching). While experienced users cope well, this represents a minor barrier for new users.

(i) Correction language is good but could improve (H2, severity 1) The verbal and text corrections are generally clear. The cosmetic-level finding suggests minor wording improvements could make corrections feel more natural and precise.

5. Conclusions

The heuristic evaluation confirms that AvatarFit has a well-designed interface that respects the majority of Nielsen's usability principles. Of the four issues identified, two are major (severity 3) and both relate to the same root cause: the absence of onboarding or pre-exercise guidance

The absence of violations in heuristics H1, H3, H4, H6, H7, H8, and H9 is particularly encouraging. It indicates that the core interaction model (skeleton feedback, exercise switching, rep counting) is intuitive and consistent.